



Chandra Lekha Guthikonda

Sr.Data Engineer

718-269-8813

gclekha6@gmail.com

<https://www.linkedin.com/in/chandra-lekha-g-657a422a1/>



Data Engineer | Hadoop | Snowflake | Python | AWS |

PROFESSIONAL SUMMARY: I'm an experienced Data Engineer with 5+ years of professional experience in developing secure, cost-efficient data solutions with a breadth of knowledge across various industries. I've developed and deployed efficient, scalable data secure, scalable, and high-performance data solutions in AWS, Snowflake, Databricks, and Apache Airflow environments. Strong background in data security, access control, compliance, and encryption for financial and enterprise-grade systems pipelines increasing customer satisfaction by 20%, streamlining integration and profiling processes by 40%, to support data enablement and analytics initiatives. Proficient in leveraging various technologies and programming languages such as Python, SQL, and Hadoop to extract, transform, and load (ETL) data for analysis and reporting purposes.

CORE SKILLS:

Cloud Platforms: AWS (S3, Redshift, Glue, Lambda, Athena, EMR, Kinesis, DynamoDB, RDS, EC2, Step Functions)

Programming & Scripting: Python, SQL, GSQL, PySpark, Shell Scripting

Big Data & Data Processing: Apache Spark, Hive, Presto, Delta Lake, Hadoop

ETL & Data Pipelines: AWS Glue, Apache Airflow, AWS Step Functions, Informatica, DBT

Streaming & Message Queues: Apache Kafka, AWS Kinesis, SQS

Databases & Data Warehousing: Amazon Redshift, Snowflake, PostgreSQL, MySQL, DynamoDB

Infrastructure as Code (IaC): Terraform, CloudFormation

Version Control & CI/CD: Git, GitHub, AWS Code Pipeline, Jenkins, Ansible

Monitoring & Logging: CloudWatch, Splunk, ELK Stack (Elasticsearch)

Security & Compliance: IAM, KMS, VPC, Data Encryption, GDPR, HIPAA

Data Governance: AWS Glue Data Catalog, Colibra, Informatica

CERTIFICATIONS:

- [DP 700: Microsoft certified Fabric Data Engineer Associate](#)
- Tiger graph Associate certification

EXPERIENCE

Bank of America

Jan 2024-Present

Data Engineer

Responsibilities:

- Designed and developed Security framework to provide fine grained access to objects in **AWS S3** using **AWS lambda**, **Dynamo DB**.
- Designed and setup enterprise data lake to provide support for various uses cases including Analytics, processing, storing and reporting of voluminous rapidly changing data.
- Implemented assessment of various AWS services Like **Amazon EMR**, **Redshift**, **S3** by optimizing **AWS Redshift** data warehouse, resulting in a 20% improvement in query performance, a 15% reduction in storage costs and enabling real-time analytics.
- Implemented the machine learning algorithms using **Python** to predict the quantity with **AWS Sage Maker**, improving predictive analytics accuracy by 25% and driving data-driven decision making.
- Involved with development of **Ansible playbooks** with **Python** and **SSH** as wrapper for management of AWS node configurations and testing playbooks on AWS instances.
- Collaborated with business teams to design **Snowflake data models**, improving data accessibility and reducing query

execution times by 30% in **Snowflake and Databricks**, reducing data processing time by **40%**.

- Designed **Snowflake-based data lakes, ETL/ELT pipelines** using **DBT, Five Tran, and Python**, and **warehouses**, optimizing performance and cost efficiency ensuring efficient data transformation and ingestion.
- Optimized **PostgreSQL indexing and partitioning strategies**, leading to **60% faster query performance**. Implemented **NoSQL solutions** using **MongoDB and DynamoDB** for semi-structured data storage and retrieval.
- Designed and optimized **complex SQL/BTEQ scripts** to extract and transform AML (Anti-Money Laundering) monitoring datasets for compliance analytics.
- Tuned long-running queries using **Teradata Explain Plans, Statistics Collection, and Partitioning** strategies, improving performance by 40%.
- Implemented **Airflow DAGs** for orchestrating **ETL workflows**, scheduling and monitoring data workflows and reducing manual intervention by 80%.
- Implemented a real-time analytics solution using **Apache Airflow, PySpark, Kafka, Snowflake, and Looker** to process event-driven data, batch processing, reducing execution time by 40% with partitioning, bucketing, and caching techniques.
- Implemented **Spark Structured Streaming** for incremental data processing and reduced batch latency by 40%.
- Designed and developed robust data pipelines for AML monitoring using **Teradata 16.x, TPT, and Informatica 10.x**.
- Ingested large volumes of structured and semi-structured data into **Cloudera CDP** platform using **Hive, Impala, and PySpark**.
- Tuned complex **SQL** queries to enhance query performance and reduce processing time for data marts by 50%.
- Developed a **CI/CD** framework using **GitHub Actions, Jenkins, and Terraform** to automate data pipeline deployments.
- Designed and implemented **efficient data pipelines** utilizing **Parquet, Avro, and ORC** for optimized storage and fast retrieval in **big data ecosystems** (Hadoop, Spark, and AWS S3).
- Supported and maintained **Hadoop clusters** in production environments (Cloudera). Developed custom MapReduce programs to extract and transform raw clickstream data. Built **Hive** tables and optimized partitioning strategies for large-scale data storage.
- Designed and implemented **efficient data access patterns** using **caching strategies, partitioning, and sharding** to optimize performance across high-throughput data pipelines and analytical workloads.
- Applied **data modeling best practices** to design scalable schemas and choose appropriate data structures for both transactional and analytical systems, improving query efficiency and storage utilization.
- Tuned large-scale data systems by implementing **indexing and denormalization techniques**, resulting in a 40% reduction in query latency for key reporting datasets.
- Worked with **Apache Arrow** for high-performance in-memory data processing, enhancing cross-system data sharing with Python, Spark, and pandas workflows
- Collaborated with analytics teams to deliver high-quality data models and reports. Implemented monitoring alerts for **HDFS** and job health using **Splunk and Grafana**.
- Developed and optimized **data transformation pipelines** using **Python**, with a strong focus on **DataFrame manipulation** via **Pandas** and **Polars**, handling large-scale datasets efficiently for analytical and ML-driven use cases.
- Engineered DAG-based workflows using **Apache Airflow** and **Dagster**, enabling modular, version-controlled pipelines for end-to-end data ingestion, transformation, and delivery.
- Designed and implemented transformation layers using **dbt (Data Build Tool)**, standardizing data models and building maintainable, testable SQL-based transformations for downstream analytics teams.
- Implemented monitoring and alerting within orchestrated pipelines to ensure data quality and freshness, reducing SLA violations by 40%.
- Automated infrastructure deployment using **Ansible playbooks** for provisioning and configuring EC2 instances, setting up web servers, and installing packages across environments.
- Actively participated in **Agile** ceremonies such as daily stand-ups, sprint planning, sprint reviews, and retrospectives as a member of a Scrum team.

Key Achievements:

- Design and Supported a serverless data pipeline using **AWS Lambda and Kinesis**, reducing data processing time by 40% and cutting operational costs by 25%.
- Automated data ingestion and transformation workflows using **Apache Airflow** and **AWS Lambda**, reducing data processing times by 30%.
- Administered a fleet of **MySQL and PostgreSQL** databases with 99.99% uptime, ensuring high availability for critical business applications.

Amtrust Financial
Data Engineer

May 2021 – Apr 2022

Responsibilities:

- Build efficient ETL (Extract, Transform, Load) pipelines using **AWS Glue, Lambda, and EMR** to process large datasets

from various sources and leveraged **AWS S3** for data storage, ensuring the optimal configuration of data lakes with partitioning and compression strategies.

- Integrated data from various on-premises and cloud sources, transforming and loading it into **AWS Redshift, Snowflake, or DynamoDB** for analysis.
- Implement **data cleansing, normalization, and aggregation** using **Python, SQL, and AWS Glue** to ensure high-quality data for reporting and analytics.
- Developed distributed data processing workflows using **PySpark** on **AWS EMR**, improving data transformation speed by 40%.
- Built real-time fraud detection pipelines leveraging **Kinesis, Lambda, and DynamoDB**, enabling instant anomaly detection.
- Designed data lake architecture on **AWS S3** with partitioning and compression, significantly reducing storage and query costs.
- Developed RESTful APIs using **AWS Lambda and API Gateway**, enabling scalable, serverless data access for analytics and reporting.
- Implemented **ETL pipelines to feed analytics dashboards**, processing millions of daily records in **Snowflake and BigQuery**.
- Built ETL pipelines using **Informatica PowerCenter and BTEQ scripts** to load patient and clinical data from various source systems to Teradata DW.
- Integrated **Hadoop Hive datasets** into Teradata using **Informatica Big Data Edition** and shell wrappers.
- Designed and implemented high-performance **Spark-based ETL** jobs to ingest and transform 10TB+ of data daily from heterogeneous sources. Migrated legacy **MapReduce workflows to Spark**, reducing execution time by over 60%.
- Integrated **Kafka** for real-time streaming pipelines and developed Spark Streaming applications to enrich and store data in **HDFS and Hive**.
- Developed optimized **Hive queries and tables** (ORC/Parquet with partitioning & bucketing). Maintained production Hadoop clusters, monitoring performance, and resolving bottlenecks and outages. Scheduled and monitored batch ETL jobs via **Autosys**, reducing job failure rate by 25% after implementing error handling logic.
- Designed and implemented end-to-end data pipelines using **Databricks and Snowflake**, processing terabytes of structured and semi-structured data.
- Containerized **Python** applications with Docker and deployed them on **AWS EC2 and Kubernetes**. Designed and implemented secure authentication layers for APIs using **AWS IAM roles and JWT** tokens.
- Developed scalable data pipelines using **Apache Airflow and PySpark** to process terabytes of structured and semi-structured data.
- Wrote complex **SQL queries** to transform raw data into actionable insights, supporting data-driven decision-making across business units.
- Automated infrastructure provisioning using **Terraform and CloudFormation**, improving **deployment efficiency by 40%**
- Developed **event-driven architectures** using **Kafka, Kinesis, and AWS Lambda** to process millions of transactions per second.
- Developed **serverless ETL pipelines** using **AWS Lambda, S3, and RDS, AWS Glue** to process large-scale financial data cutting infrastructure costs by **30%**.
- Developed and maintained scalable REST APIs using **Spring Boot** and **Java 11** for e-commerce and logistics platforms.
- Integrated **PostgreSQL and MongoDB** databases with ORM tools such as Hibernate and JPA.
- Migrated a monolithic legacy system to a **microservices** architecture deployed on **Kubernetes (EKS)**.
- Developed a custom **Python library** for automated data quality checks and anomaly detection, reducing manual data validation efforts by 80% and improving data integrity across 50+ critical dataset.
- Implemented partitioning and bucketing strategies to improve **query performance** with **Oozie and Airflow** for automating **Hive** data processing.
- Created reusable **Ansible roles** and playbooks to manage configurations and streamline server onboarding across QA, staging, and production. Developed **custom Ansible modules** to handle specialized internal automation requirements. Used **Ansible** for managing Linux configurations, setting up users, updating software, and configuring firewalls across hundreds of virtual machines.

Key Achievements:

- Streamlined data storage solutions by migrating legacy systems to **AWS RDS**, achieving a 50% improvement in data retrieval speeds and enhancing system reliability.
- Helped in the deployment of a cloud-based data warehouse on **AWS Redshift**, supporting business intelligence initiatives and improving reporting capabilities by 40%.

Junior Data Engineer

Responsibilities:

- Analyzed large and critical datasets using **SQL**, **Python**, **Cloudera**, **HDFS**, **Pig**, **Sqoop** and **Spark**.
- Developed a real-time dashboard using **Python** and **SQL** to monitor website traffic and user behavior for an patients record platform by Integrating data from web servers, databases, and third-party APIs to provide up-to-the-minute insights on key performance indicators such as website traffic, medication outages, and revenue.
- Collaborated with the DevOps team to containerize **PySpark** applications using **Docker** and deployed them in **Kubernetes** clusters for scalable and reliable execution.
- Designed and implemented **ETL pipelines** in **Python**, using libraries like **Pandas**, **NumPy**, and **PySpark** to handle data extraction, transformation, and loading from various sources (e.g., **SQL databases**, **APIs**, **CSV**, **JSON**).
- Automated ETL workflows using **Apache Airflow**, orchestrating the integration of **AWS S3**, **Redshift**, and other data sources for batch and streaming pipelines.
- Developed and managed **ETL jobs** using **AWS Glue** to process large-scale datasets stored in **Amazon S3**.
- Built and orchestrated real-time data workflows using **Airflow** and **Kafka**, enabling on-time data delivery and monitoring for financial reporting.
- Developed and maintained **data transfer solutions** using **SFTP**, **FTP**, and **API-based** integrations across on-prem and cloud systems.
- Developed and supported data pipelines using **AWS Serverless** technologies, including **Lambda**, **S3**, and **API Gateway**, to automate data ingestion from external APIs.
- Designed and optimized **complex SQL queries** for data extraction, aggregation, and reporting in **PostgreSQL** databases.
- Optimized **SQL queries and data models**, reducing dashboard load time by 60% for large datasets in **Redshift** and **Snowflake**.
- Developed **data processing pipelines** using **Apache Spark on AWS EMR**, improving batch processing efficiency.
- Migrated **on-prem ETL workflows** to **AWS Airflow on EKS**, reducing processing time by 30%.
- Configured **AWS Glue**, **Snowflake**, and **Databricks** to read and write **Parquet** and **Avro** for efficient data warehousing and analytics.
- Managed **VPC networking** for secure data movement across cloud and on-prem systems.
- Hands-on experience with **JSON-based data models**, including healthcare-specific standards like **FHIR (Fast Healthcare Interoperability Resources)**, for managing semi-structured patient data in compliance with interoperability standards.
- Collaborated with architecture teams to define **serialization and storage strategies**, balancing performance, scalability, and compliance needs across diverse data domains.
- Designed **CDC-based ingestion pipelines** using **FiveTran** and **AWS DMS**, enabling near real-time data replication.
- Built **self-service ETL frameworks** using **Argo Workflows** and **AWS Step Functions**, improving automation and reducing operational overhead. Participated in **Daily stand-up Agile/Scrum meetings** to discuss the enhancements to be made for the existing requirements.

Key Achievements:

- Automated data cleansing and validation processes using **SQL** stored procedures and functions, reducing manual intervention and improving data quality by 25%.
- Optimized costs and improved performance by executing partitioning, caching, and job parallelism in **PySpark** jobs on **Azure Databricks**.
- Enforced performance tuning strategies for **SQL queries** used in a real-time patient monitoring system through query execution plans and optimized indexes, and rewrote complex queries to improve efficiency by 36%.
- Applied unit testing coverage to improve code efficiency using **Pytest**.

EDUCATIONAL DETAILS:

Masters in Computer Science
University of Central Missouri

May 2022 – Dec 2023

Bachelors in Electronics and Communications
CMR Institute of Technology

Aug 2016 - Aug 2020